

Advies (voorbehouden aan de Ethische Commissie)/Advice (for Ethical Commission):

Handtekening Voorzitter Ethische Commissie:



22/4/2022

☐ gunstig

☒ gunstig nadat het werd aanpast of vragen werden beantwoord

☐ ongunstig

Opmerkingen

Aanvraag/Application:

Projectnr.	P069/2022
Aangevraagd door	Zsuzsanna Vegh (u0048902) op 2021-01-06 14:29:31
Status	Bij ECD (op afdrukdatum 2022-04-22 08:28:25)
Pijnscore ECD	ernstig
Pijnscore (Post-experimentele evaluatie)	Nog niet geregistreerd?

Gebruikers

Afdeling / Labo	OG Labo voor Biologische Psychologie
Erkenningsnummer	LA1210517
Wordt (een deel van) de proef uitgevoerd in andere instelling?	Neen

Personeel

Projectverantwoordelijke	Zsuzsanna Vegh (u0048902)				
Opleidingsverantwoordelijke	Zsuzsanna Vegh (u0048902)				
Personeel dat aan de proeven deelneemt	Familienaam	Voornaam	Diploma	Functie	Status
	D'Hooge	Rudi	MSc biology-zoology, MSc & PhD research psychology, PhD biomed sci	Proefleider (PI)	OK
	Vegh	Zsuzsanna	Ms Pharmacy, PhD biomedical sciences	Actieve medewerker	OK
	Craeghs	Livine	master in biologie	Actieve medewerker	OK
	Ossorio Salazar	Victoria Aurora	PhD student	Actieve medewerker	OK
	Moonen	Louise	laborant	Actieve medewerker	OK

Project

Titel	Onderzoek van de therapeutische mogelijkheden voor cognitieve stoornissen geïnduceerd door chemotherapie (Therapeutic avenues for chemotherapy induced cognitive impairment)
Startdatum	01/04/2022
Einddatum	31/03/2027
Verbonden aan projectnr.	P068/2020
Bijkomende informatie	This project builds on the model established in 068/2020 where methotrexate was used as a model for chemotherapy induced cognitive impairment. This project will

Beknopte beschrijving van het onderzoeksproject	expand to clinically relevant models. Leukemia is one of the most frequent childhood cancer with a high fatality rate when left untreated. However, progress in therapeutic solutions increased survival and cure rate of children diagnosed with childhood leukemia to over 80%. One of the long-term complications are behavioral and cognitive problems observed later in life. We are interested in identifying novel interventions that might benefit childhood cancer survivors by improving their behavioral and cognitive problems. Methotrexate is one of the most frequently use chemotherapeutics in childhood leukemia. In addition to its effect on cancer cells, it also interferes with neural development and oligodendritic myelination (OLm), a process that is very active during early childhood CNS maturation. We have established a methotrexate (MTX) model in mouse with robust disruption of OLm resulting in cognitive impairment (amendment of 068/2020). In this follow up project we want to expand our model using the clinically relevant hyper-CVAD cocktail, consisting of several chemotherapeutics and used in juvenile ALL patients.		
Domein van het onderzoek	Translationeel of toegepast onderzoek		
Verduidelijking domein	The goal is to establish a murine model close to the clinically used Hyper-CVAD treatment, and testing possible therapeutic interventions.		
Vereisten wetgeving	n.v.t.		
Alternatieve methoden	Since we are looking at changes in behavior and improvement of cognition, we will be using live animals. However, we have been optimizing our testing protocols to test one animal in a well defined series of tests to reduce the required number of animals to a minimum.		
Goedgekeurde bronnen	Goedgekeurde bronnen	Specifieer	Geraadpleegd op
	ONDERZOEK		
	PubMed		01/03/2022
	PREPARE guidelines Norocopa(https://norecopa.no/PREPARE)		01/03/2022
Inschatting volgens Vlare van de risicoklasse van het dierexperimentele werk	1		
Gebruikt u Tamoxifen?	neen		
Gebruikt u andere chemicaliën (andere dan anesthesie)?	Methotrexate, doxorubicin, vincristine, dexamethasone, cyclophosphamide,		
Dieren			
Dieren besteld via het Proefdieren centrum?	ja		
Vrijstelling inzake gebruik vereist?	neen		
Huisvesting in het Proefdieren centrum?	Animalia in BioPSy		
Kan het project uitgevoerd worden met lagere diersoorten?	neen The readouts, such as higher order cognition and social interaction are difficult to obtain in lower organisms. Mice are in terms of anatomical, physiological, genetic and behavioral aspects the closest to clinical situation.		
Is een samenwerking met een ander laboratorium mogelijk?	neen		
Maatregelen duplicatie	We have extensively search public domains (pubmed, google scholar) for pre-existing information. This proposed project is part of a recently funded FWO project. We have found no information regarding this specific cocktail of medication.		
Dierproeven			
Totaal aantal dieren over alle dierproeven	Diersoort	Aantal	
	muis	192	
Dierproef 1			
Titel	Pilot: establishing a clinical relevant Hyper-CVAD model		
Type experiment	Pilot		
Klasse	Ernstig		

Inschatting van de ernst van de dierproef	We will administer a cocktail of chemotherapeutics which will affect the well being of the animals. We target a dose that is not lethal for the animals, but they will certainly feel sick.					
Dieren	Soort	Stam/ras	Aantal	Genetische status	Ontwikkeling sstadium	Geslacht
	muis	C57BL/6J	12	GS1	PND21-PND70	Geen voorkeur
Korte motivatie voor deze dierproef (wat wil u leren van dit experiment)/de gebruikte stammen/het gekozen geslacht?	We will establish a working hyper-CVAD model using a chemotherapeutic cocktail, but we certainly need to see if the doses calculated based on clinical doses are viable in mouse. We will start with standard doses that can be found in literature (either from single dose in mouse or from doses used in clinic), and will follow a strict dosing scheme. If we observe fatality, we will reduce the amount of chemotherapeutics.					
Aantal groepen en aantal dieren per groep	We will use only one group, and apply the established hyper-CVAD treatment schedule. We will start with n=3, and based on the observation, we will decide whether to decrease or maintain the chosen doses. To confirm the correct dose 3 additional animals will be treated. From previous experiments using methotrexate alone, we observed a 15% fatality rate. We will therefore include 3 animals initially, hoping that we have at least 2 animals that will survive as a criterion for dosing choice (maximum 30% fatality rate). In case we observe a higher fatality rate than 30%, we will reduce the doses by 30%.					
Verantwoording aantal						
Toelichting bij de geplande ingrepen op het levende dier - beschrijf chronologisch de geplande manipulaties	PND21: mice will be weaned, weighed and ear tagged. PND21: injection of Methotrexate i.p. (100mg/kg) PND22-PND27: daily check for normal development (weight gain, activity, normal movement) PND28: injection of Methotrexate i.p. (100mg/kg) PND29: start of HyperCVAD regimen: administration of Cyclophosphamide (CY, 10mg/kg) and Vincristine (VI, 0.5mg/kg) i.p., monitoring development: weight gain, activity, normal movement (at 8am, and at 6pm) PND30: administration of CY (10mg/kg) and VI (0.5mg/kg) i.p., monitoring development: weight gain, activity, normal movement (at 8am, and at 6pm) PND31: administration of doxorubicin (10mg/kg) i.p. monitoring development: weight gain, activity, normal movement (at 8am, and at 6pm) PND32: administration of dexamethasone (5mg/kg) i.p. monitoring development: weight gain, activity, normal movement (at 8am, and at 6pm) PND33-34: monitoring development: weight gain, activity, normal movement (once per day) PND35: repeat from PND28-34 PND42: repeat from PND28-34 PND49: limited behavioral testing including 1) Rotarod: animals are tested for motor balance on accelerated rotating rod (5-40rpm over 5 min) 2) Open field: animals are placed in an open field and spontaneous exploration is recorded for 10 min using a overhead camera and tracking software 3) Elevated plus maze: For 10 min animals are placed on an elevated maze in form of a plus. Time in closed and open arm is recorded and tracked. 4) Social approach in 3 chamber test: for 25 min animals are tested in the SPSN test , where approach to unknown stranger mouse is recorded using overhead cameras. 5) Pain sensitivity via tail flick test: animals are restrained and the tail is dipped in warm water (45-48-50 degree C, 10 min in between temperatures), and latency to remove tail is recorded. A maximal latency of 60s is given. Each of these tests are done on consecutive days (only one test per day). After behavioral testing, animals are killed by transcardial perfusion under deep pentobarbital (100mg/kg) sedation. Tissues are collected for further analysis. If of the initial n=3 animals at least 2 survive and show growth, we will add another n=3 to confirm our findings before starting on DP2. If 2 animals of the initial n=3 die during treatment, we will reduce the amount of chemotherapeutics by 30%, and test another group of n=3 (see schema). We will include maximally 2 dose reduction steps (maximally 12 animals). If we still see fatality, we will reevaluate the model and resubmit a new project.					
Hergebruik	neen					
Verantwoordelijke voor de follow-up van het dierenwelzijn	Zsuzsanna Vegh (PhD, lab manager, daily also on weekends), Louise Moonen (lab technician)					
Wijze waarop de follow-up van het	We expect that the animals will become sick from the treatment. We will monitor					

dierenwelzijn tijdens de dierproef gewaarborgd wordt	closely (daily or twice a day) the animals with regards to body weight, normal movement, activity and pain (using the mouse pain scale). We will provide extra food (peanut butter and softened food pellets) and easier access to water (longer nipples) to reduce suffering. If we observe a drastic drop in well-being (reduction of body weight by 10-15% within 24h, no activity, hunched backs), we will first give glucose/saline (5% body weight) and observe animal for 4h. If condition is not improving, we will euthanize the animals. This will be used as information to lower the dose administered.
Methoden gebruikt om ongemak te verminderen/uit te schakelen	We will provide extra food (peanut butter) and easier access to water (longer nipples). We will administer glucose (5% in saline, i.p.) when dehydration is observed.
Lijst van medicatie met dosis, toedieningsweg, duur en frequentie	Methotrexate 4x (PND21,28,35,42), 100 mg/kg i.p. Cyclophosphamide 6x (PND29, 30, 36, 37, 43, 44) 10mg/kg i.p. Vincristine 6x (PND29, 30, 36, 37, 43,44) 0.5mg/kg i.p. Doxorubicin 3x (PND31, 38, 45) 10mg/kg, i.p. Dexamethasone 3x (PND32, 39, 46) 5mg/kg, i.p.
Verfijning huisvestigings- en verzorgingsomstandigheden	Animals will be group housed, with extra nest material, small paper rolls for hiding, and peanut butter
Humane eindpunten specifiek voor dit project, experiment of diermodel	24h after injection: - loss of bodyweight >15% - low activity (hunched in corner, non-responsive) - no improvement with 5% bw injection of glucose/saline
Beschrijf pilootstudie	
Preciseer per diersoort de voorziene methode(n) om het dier te doden en de dood te bevestigen	Between PND21-49: CO2 inhalation and removal of head After behavioral testing: deep sedation with pentobarbital (100mg/kg), an transcardial perfusion
Preciseer de betrokken dieren en hun bestemming	n/a
Adoptie of vrijlating	n/a

Dierproef 2

Titel	Behavioral phenotyping of a clinical relevant Hyper-CVAD model					
Type experiment	Experiment					
Klasse	Ernstig					
Inschatting van de ernst van de dierproef	We will administer a cocktail of chemotherapeutics which will affect the well being of the animals. We target a dose that is not lethal for the animals, but they will certainly feel sick.					
Dieren	Soort	Stam/ras	Aantal	Genetische status	Ontwikkeling stadium	Geslacht
	muis	C57BL/6J	30	GS1	PND21-PND90	Mannelijk
	muis	C57BL/6J	30	GS1	PND21-PND90	Vrouwelijk
Korte motivatie voor deze dierproef (wat wil u leren van dit experiment)/de gebruikte stammen/het gekozen geslacht?	Based on the injection protocol we established in DP1 pilot, we will compare the effect of Hyper-CVAD model to vehicle treated animals in a series of behavior. Since we will test the animals in a well-defined battery of several tests, we can reduce the numbers of animals needed to evaluate behavioral changes. We will include both males and females, by including the entire nest (bred in our animalium) and not only one sex.					
Aantal groepen en aantal dieren per groep	We will have per sex 2 groups: Control: n=15 Males; n=15 Females Hyper-CVAD: n=15 Males; n=15 Females We will compare control versus Hyper-CVAD.					
Verantwoording aantal	We consider a difference between controls and Hyper-CVAD of 30% in any readout of our behavioral battery as biological relevant and as an indication of group difference (compared in a two-sided, unpaired t-test). Considering a shared SD of 20-25%, and accepting an alpha=0.05, and power of 80%, we calculate that we need n=12/group. Due to an expected fatality rate of 10-20% we will include n=15/group.					
Toelichting bij de geplande ingrepen	The doses of all the chemotherapeutics will be established in DP1 pilot.					

op het levende dier - beschrijf
chronologisch de geplande
manipulaties

PND21: mice will be weaned, weighed and ear tagged.
PND21: injection of Methotrexate i.p. (max. 100mg/kg/ or DP1)
PND22-PND27: daily check for normal development (weight gain, activity, normal movement)
PND28: injection of Methotrexate i.p. (max 100mg/kg/ or DP1)
PND29: start of HyperCVAD regimen: administration of Cyclophosphamide (CY, max 10mg/kg/ or DP1) and Vincristine (VI, max 0.5mg/kg/ or DP1) i.p., monitoring development: weight gain, activity, normal movement (at 8am, and at 6pm)
PND30: administration of CY (max 10mg/kg/ or DP1) and VI (max 0.5mg/kg / or DP1) i.p., monitoring development: weight gain, activity, normal movement (at 8am, and at 6pm)
PND31: administration of doxorubicin (max 10mg/kg/ or DP1) i.p. monitoring development: weight gain, activity, normal movement (at 8am, and at 6pm)
PND32: administration of dexamethasone (max 5mg/kg/ or DP1) i.p. monitoring development: weight gain, activity, normal movement (at 8am, and at 6pm)
PND33-34: monitoring development: weight gain, activity, normal movement (once per day)
PND35: repeat from PND28-34
PND42: repeat from PND28-34
PND49: limited behavioral testing including
1) Rotarod: animals are tested for motor balance on accelerated rotating rod (5-40rpm over 5 min)

2) Open field: animals are placed in an open field and spontaneous exploration is recorded for 10 min using a overhead camera and tracking software

3) Elevated plus maze: For 10 min animals are placed on an elevated maze in form of a plus. Time in closed and open arm is recorded and tracked.

4) Social approach in 3 chamber test: for 25 min animals are tested in the SPSN test , where approach to unknown stranger mouse is recorded using overhead cameras.

5) T-maze working memory: Animals are placed 2x for 10 min on an elevated maze in the form of a T. In the first trial, one goal arm is closed off by a barrier, preventing exploration. In the second trial, both goal arms are freely accessible and time spent in each arm is recorded.

6) Pain sensitivity via tail flick test: animals are restrained and the tail is dipped in warm water (45-48-50 degree C, 10 min in between temperatures), and latency to remove tail is recorded. A maximal latency of 60s is given.

7) Spatial learning in Morris water maze: Animals are trained for 2*5d to locate a hidden platform in a circular pool. Maximum trial duration is 100s, water temp is 26+/- 1C to prevent hyperthermia. On day 6 and 11, spatial reference memory is evaluated in 100s long probe trials, where the platform was removed.

8) Passive avoidance learning: On day 1, dark-habituated animals are placed in a brightly illuminated box with access to a dark box. When the animal enters the dark box, the door is closed and a short mild foot shock (0.3mA, 2s) is applied. After 20s. the animal is removed and placed back in its home cage. 24h later, the mouse is placed again in the light box, and latency to enter dark the dark box is recorded (with a maximum cut-off of 5 min). No foot shocks are delivered.

Each of these tests are done on consecutive days (only one test per day).
After behavioral testing, animals are killed by transcardial perfusion under deep pentobarbital (100mg/kg) sedation. Tissues are collected for further analysis.

Hergebruik	neen
Verantwoordelijke voor de follow-up van het dierenwelzijn	Zsuzsanna Vegh (PhD, lab manager, daily also on weekends), Louise Moonen (lab technician)
Wijze waarop de follow-up van het dierenwelzijn tijdens de dierproef gewaarborgd wordt	We expect that the animals will become sick from the treatment. We will monitor closely (daily or twice a day) the animals with regards to body weight, normal movement, activity and pain (using the mouse pain scale). We will provide extra food (peanut butter and softened food pellets) and easier access to water (longer nipples) to reduce suffering. If we observe a drastic drop in well-being (reduction of body weight by 10-15% within 24h, no activity, hunched backs), we will first give glucose/saline (5% body weight) and

	observe animal for 4h. If condition is not improving, we will euthanize the animals. This will be used as information to lower the dose administered.
Methoden gebruikt om ongemak te verminderen/uit te schakelen	We will provide extra food (peanut butter) and easier access to water (longer nipples). We will administer glucose (5% in saline, i.p.) when dehydration is observed.
Lijst van medicatie met dosis, toedieningsweg, duur en frequentie	Methotrexate 4x (PND21,28,35,42), 100 mg/kg i.p. Cyclophosphamide 6x (PND29, 30, 36, 37, 43, 44) 10mg/kg i.p. Vincristine 6x (PND29, 30, 36, 37, 43,44) 0.5mg/kg i.p. Doxorubicin 3x (PND31, 38, 45) 10mg/kg, i.p. Dexamethasone 3x (PND32, 39, 46) 5mg/kg, i.p.
Verfijning huisvestigings- en verzorgingsomstandigheden	Animals will be group housed, with extra nest material, small paper rolls for hiding, and peanut butter
Humane eindpunten specifiek voor dit project, experiment of diermodel	24h after injection: - loss of bodyweight >15% - low activity (hunched in corner, non-responsive) - no improvement with 5% bw injection of glucose/saline within 4h
Beschrijf pilootstudie	
Preciseer per diersoort de voorziene methode(n) om het dier te doden en de dood te bevestigen	Between PND21-49: CO2 inhalation and removal of head After behavioral testing: deep sedation with pentobarbital (100mg/kg), an transcardial perfusion
Preciseer de betrokken dieren en hun bestemming	n/a
Adoptie of vrijlating	n/a

Dierproef 3

Titel	Behavioral rescue of cognitive impairment of Hyper-CVAD model					
Type experiment	Experiment					
Klasse	Ernstig					
Inschatting van de ernst van de dierproef	We will administer a cocktail of chemotherapeutics which will affect the well being of the animals. We target a dose that is not lethal for the animals, but they will certainly feel sick.					
Dieren	Soort	Stam/ras	Aantal	Genetische status	Ontwikkeling sstadium	Geslacht
	muis	C57BL/6J	60	GS1	PND21-PND90	Vrouwelijk
	muis	C57BL/6J	60	GS1	PND21-PND90	Mannelijk
Korte motivatie voor deze dierproef (wat wil u leren van dit experiment)/de gebruikte stammen/het gekozen geslacht?	We will evaluate, if controlled social and environmental enrichment (MARLAU cages) will alleviate the cognitive impairment induced by Hyper-CVAD treatment. This manipulation could be viewed to correspond to increased tutoring, social and psychological support and possibly psychosocial coaching in a clinical setting. Since we will test the animals in a well-defined battery of several tests, we can reduce the numbers of animals needed to evaluate behavioral changes. We will include both males and females, by including the entire nest (bred in our animalium) and not only one sex.					
Aantal groepen en aantal dieren per groep	We will have per sex 4 groups: In Marlau (social & environmental enriched cages, large groups n=7-8) Control: n=15 Hyper-CVAD: n=15 In traditional cages (minimal enrichment, small groups n=3): Control: n=15 Hyper-CVAD: n=15 total n=60/sex					
Verantwoording aantal	We consider a difference between controls and Hyper-CVAD of 30% in any readout of our behavioral battery as biological relevant and as an indication of group difference (compared in a two-sided, unpaired t-test). Considering a shared SD of 20-25%, and					

Toelichting bij de geplande ingrepen op het levende dier - beschrijf chronologisch de geplande manipulaties

accepting an $\alpha=0.05$, and power of 80%, we calculate that we need $n=12/\text{group}$. Due to an expected fatality rate of 10-20% we will include $n=15/\text{group}$.

The doses of all the chemotherapeutics will be established in DP1 pilot.

PND21: mice will be weaned, weighed and ear tagged. Males and females will be separated, but placed in groups of either 7-8 or 3 (see groups). All animals will still be housed in standard cages to allow easy access to water and food.

PND21: injection of Methotrexate i.p. (max. 100mg/kg/ or DP1)

PND22-PND27: daily check for normal development (weight gain, activity, normal movement).

PND28: injection of Methotrexate i.p. (max 100mg/kg/ or DP1)

PND29: start of HyperCVAD regimen: administration of Cyclophosphamide (CY, max 10mg/kg/ or DP1) and Vincristine (VI, max 0.5mg/kg/ or DP1) i.p., monitoring development: weight gain, activity, normal movement (at 8am, and at 6pm)

PND30: administration of CY (max 10mg/kg/ or DP1) and VI (max 0.5mg/kg / or DP1) i.p., monitoring development: weight gain, activity, normal movement (at 8am, and at 6pm)

PND31: administration of doxorubicin (max 10mg/kg/ or DP1) i.p. monitoring development: weight gain, activity, normal movement (at 8am, and at 6pm)

PND32: administration of dexamethasone (max 5mg/kg/ or DP1) i.p. monitoring development: weight gain, activity, normal movement (at 8am, and at 6pm)

PND33-34: monitoring development: weight gain, activity, normal movement (once per day)

PND35: placing in Marlau cages (or leave in standard cages) in groups of 7-8 animals (maximal cage occupancy is 21-16 animals)

PND35: repeat injections from PND28-34

PND42: repeat injections from PND28-34

PND63: Animals are moved from Marlau to regular cages during behavioral testing, but are moved back after testing each day.

Limited behavioral testing includes

- 1) Rotarod: animals are tested for motor balance on accelerated rotating rod (5-40rpm over 5 min)
- 2) Open field: animals are placed in an open field and spontaneous exploration is recorded for 10 min using a overhead camera and tracking software
- 3) Elevated plus maze: For 10 min animals are placed on an elevated maze in form of a plus. Time in closed and open arm is recorded and tracked.
- 4) Social approach in 3 chamber test: for 25 min animals are tested in the SPSN test , where approach to unknown stranger mouse is recorded using overhead cameras.
- 5) T-maze working memory: Animals are placed 2x for 10 min on an elevated maze in the form of a T. In the first trial, one goal arm is closed off by a barrier, preventing exploration. In the second trial, both goal arms are freely accessible and time spent in each arm is recorded.
- 6) Pain sensitivity via tail flick test: animals are restrained and the tail is dipped in warm water (45-48-50 degree C, 10 min in between temperatures), and latency to remove tail is recorded. A maximal latency of 60s is given.
- 7) Spatial learning in Morris water maze: Animals are trained for 2*5d to locate a hidden platform in a circular pool. Maximum trial duration is 100s, water temp is 26+/- 1C to prevent hyperthermia. On day 6 and 11, spatial reference memory is evaluated in 100s long probe trials, where the platform was removed.
- 8) Passive avoidance learning: On day 1, dark-habituated animals are placed in a brightly illuminated box with access to a dark box. When the animal enters the dark box, the door is closed and a short mild foot shock (0.3mA, 2s) is applied. After 20s. the animal is removed and placed back in its home cage. 24h later, the mouse is placed again in the light box, and latency to enter dark the dark box is recorded (with a maximum cut-off of 5 min). No foot shocks are delivered.

Each of these tests are done on consecutive days (only one test per day). After behavioral testing, animals are killed by transcardial perfusion under deep pentobarbital (100mg/kg) sedation. Tissues are collected for further analysis.

Hergebruik

neen

Verantwoordelijke voor de follow-up van het dierenwelzijn	Zsuzsanna Vegh (PhD, lab manager, daily also on weekends), Louise Moonen (lab technician)
Wijze waarop de follow-up van het dierenwelzijn tijdens de dierproef gewaarborgd wordt	We expect that the animals will become sick from the treatment. We will monitor closely (daily or twice a day) the animals with regards to body weight, normal movement, activity and pain (using the mouse pain scale). We will provide extra food (peanut butter and softened food pellets) and easier access to water (longer nipples) to reduce suffering. If we observe a drastic drop in well-being (reduction of body weight by 10-15% within 24h, no activity, hunched backs), we will first give glucose/saline (5% body weight) and observe animal for 4h. If condition is not improving, we will euthanize the animals. This will be used as information to lower the dose administered.
Methoden gebruikt om ongemak te verminderen/uit te schakelen	We will provide extra food (peanut butter) and easier access to water (longer nipples). We will administer glucose (5% in saline, i.p.) when dehydration is observed.
Lijst van medicatie met dosis, toedieningsweg, duur en frequentie	Methotrexate 4x (PND21,28,35,42), 100 mg/kg i.p. Cyclophosphamide 6x (PND29, 30, 36, 37, 43, 44) 10mg/kg i.p. Vincristine 6x (PND29, 30, 36, 37, 43,44) 0.5mg/kg i.p. Doxorubicin 3x (PND31, 38, 45) 10mg/kg, i.p. Dexamethasone 3x (PND32, 39, 46) 5mg/kg, i.p.
Verfijning huisvestigings- en verzorgingsomstandigheden	Animals will be group housed, with extra nest material, small paper rolls for hiding, and peanut butter
Humane eindpunten specifiek voor dit project, experiment of diermodel	24h after injection: - loss of bodyweight >15% - low activity (hunched in corner, non-responsive) - no improvement with 5% bw injection of glucose/saline within 4h
Beschrijf pilootstudie	
Preciseer per diersoort de voorziene methode(n) om het dier te doden en de dood te bevestigen	Between PND21-49: CO2 inhalation and removal of head After behavioral testing: deep sedation with pentobarbital (100mg/kg), an transcardial perfusion
Preciseer de betrokken dieren en hun bestemming	n/a
Adoptie of vrijlating	n/a

Financieel

Financiële antenne	An Van Kets (u0057838)
SAP debiteur	
Krediettype	KULC
Toegevoegde financieringsbron(nen)	Er werden nog geen financiële bronnen toegevoegd.
Europees project	neen